

File Permissions/Mode

- File permission types: r (read), w (write), x (execute)
- Permission levels: u (user), g (group), o (other)
- File mode: u-rwx g-rwx o-rwx
- Example: hello.txt -- user can read/write and execute, group can read and execute, other can only read.
 - hello.txt mode: u=rwx g=r-x o=r--
 - rwx r-x r--
 - 111 101 100 = 754
 - chmod 754 hello.txt

chmod

- chmod permissions filename
- Permissions:
 - +x/r/w - add permission to all
 - -x/r/w - remove permission of all
 - u+x/r/w - add permission to user
 - u-x/r/w - remove permission of user
 - g+x/r/w - add permission to group
 - g-x/r/w - remove permission of group
 - o+x/r/w - add permission to others
 - o-x/r/w - remove permission of others
 - 0774 - apply rwx to user, rwx to group and r to others

chown

- sudo chown username filename
- sudo chown :groupname filename
- sudo chown username:groupname filename

VI Editor

- sudo apt-get install vim
- VI editor works in two modes
 - command mode
 - insert mode
- press i - to go into insert mode
- press Esc - to go into command mode
- VI editor commands:
 - :w - write/save into file
 - :q - quit vi editor

- :wq - save and quit
- :y - to copy current line
- :ny - to copy nth line
- yy - to copy current line
- nyy - copy n lines from current line
- :m,ny - copy from mth line to nth line
- :d - to cut current line
- :nd - to cut nth line
- dd - to cut current line
- ndd - cut n lines from current line
- :m,nd - cut from mth line to nth line
- press p - to paste copied line on next line of current line
- yw - copy from current position upto next word
- yiw - copy current word
- y\$ - copy from cursor position upto end of line
- y^ - copy from cursor position upto start of line
- vim -o - to open multiple files horizontally
- vim -O - to open multiple files vertically
- ctrl + ww - to go into next file
- /pattern - to search the pattern
- n - to go on next occurrence
- 🙄/pattern1/pattern2/ - find and replace only first occurrence of current line
- 🙄/pattern1/pattern2/g - find and replace all occurrences of current line
- :%s/pattern1/pattern2/g - find and replace all occurrences of file
- To do setting in vi editor
 - create file into home directory as below:

```
vim ~/.vimrc
```

- add below content into it

```
set number
set autoindent
set tabstop=4
set nowrap
```

Shell Scripting/Programming

- Shell Script - set of commands with some programming constructs, written for shell
- commands - shell commands
- Programming construct - control flow structures, functions, arrays
 - Decision control structure (if else)
 - Selection control structure (case)
 - Loop control structure (while, until, for)
- Shell scripts are interpreted by shell

Applications

- Shell scripts are used by administration people for installations, setups, automations
- Administrative tasks e.g. User creations, Troubleshooting, Router/Firewall config on commandline, etc.
- Automation of tasks e.g. Cron jobs, Scripts for building softwares, etc.
- Application installers

Limitations

- Slow execution -- Interpreted. Cannot be used for performance centric applications.

Procedure to write and execute scripts

```
* Step 1 - create file with extension .sh
* $ vim script.sh
* Step 2 - execute the script
* Option 1
* $ bash script.sh

* Option 2
* $ chmod +x script.sh
* $ ./script.sh
```